

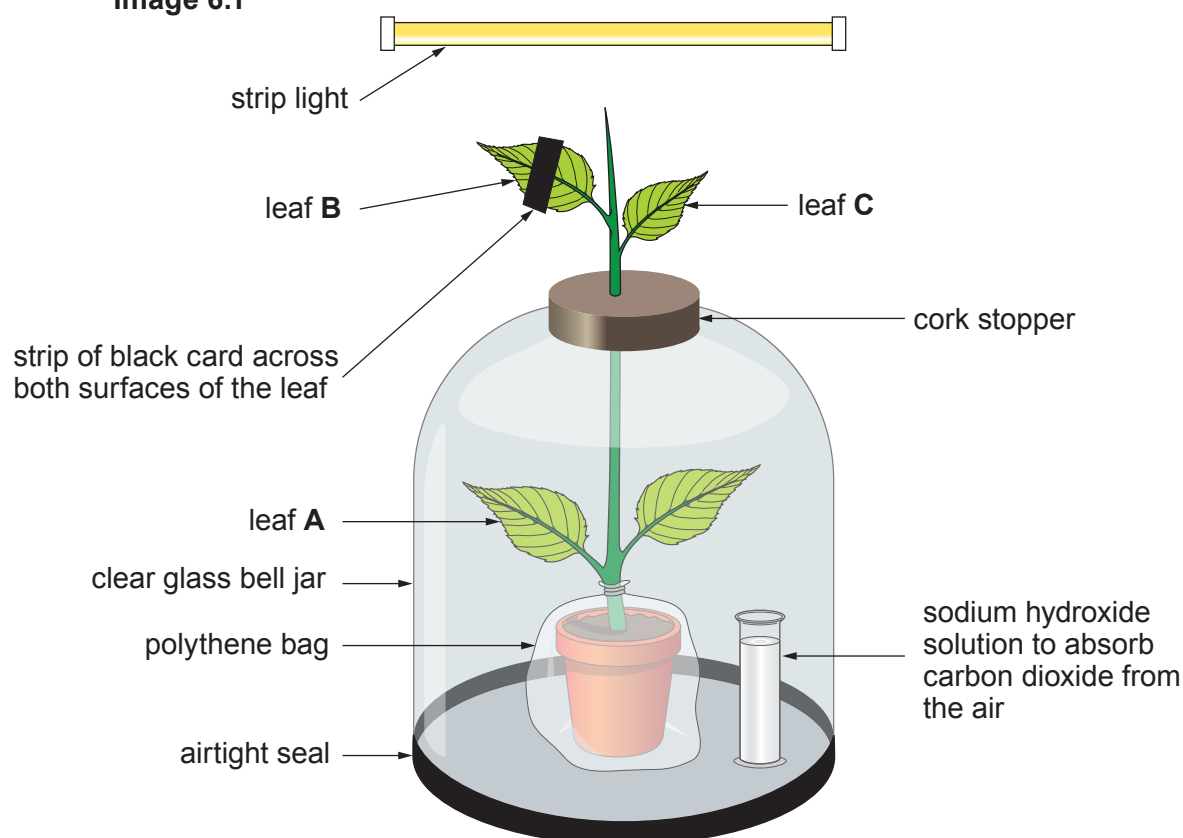
6. (a) State the function of chlorophyll.

[1]

(b) James investigated photosynthesis in a potted plant using the following method:

- Place a potted plant in the dark for 24 hours to destarch the leaves.
- After 24 hours, set up the apparatus as shown in **Image 6.1**.

Image 6.1



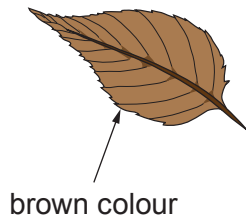
- Leave the apparatus containing the potted plant in bright light for 24 hours.
- Remove leaves **A**, **B** and **C**, then:
 - remove the card from leaf **B**
 - place the leaves in boiling water
 - place the leaves in boiling ethanol to remove the chlorophyll
 - dip the leaves in hot water and blot dry
- Carry out the starch test by dropping iodine solution onto the leaves.



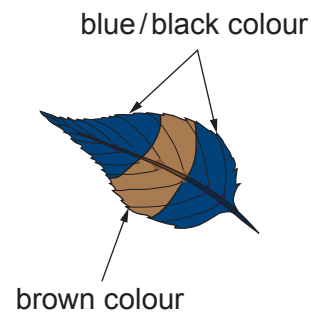
The appearance of each leaf after the starch test is shown in **Image 6.2**.

Image 6.2

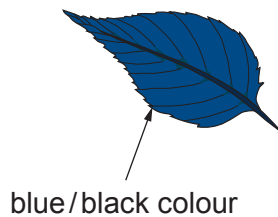
Leaf A



Leaf B



Leaf C



- (i) I. Explain the results for leaf **A** in **Image 6.2**.

[3]

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- II. Explain the appearance of the brown band across the middle of leaf **B** in **Image 6.2**.

[2]

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- (ii) Identify the control leaf (**A**, **B**, or **C**) and give the reason for your choice.

Leaf:

[1]

Reason:

[1]

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